

RESEARCH ARTICLE

Genome sequencing and molecular characterisation of XDR *Acinetobacter baumannii* reveal complexities in resistance: Novel combination of sulbactam–durlobactam holds promise for therapeutic intervention

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Abstract

Emerging nosocomial strains of *Acinetobacter baumannii* are of recent concern as they are expressing extensive drug resistance (XDR). Using whole-genome sequencing and molecular characterisation analysis, the current study reveals the presence of carbapenemase genes in 92.86% of studied Indian isolates. These included *bla*_{OXA-51}, *bla*_{OXA-23}, *bla*_{OXA-58}, and *bla*_{NDM} genes, with over a third expressing dual carbapenemase genes. As per the MLST scheme, IC2^{Oxf}/CC2^{Pas} was the predominant clone, with 57.14% isolates belonging to this lineage. The presence of these carbapenemase genes resulted in sulbactam (SUL) resistance (MIC: 16–256 µg/ml) in all of the studied isolates. The efficacy of durlobactam (DUR), a novel β-lactamase inhibitor that also inhibits PBP2 was assessed through in silico intermolecular interaction analysis. Several nonsynonymous single nucleotide polymorphisms were identified in PBP2 (G264S, I108V, S259T) and PBP3 (A515V, T526S) sequences. Minimal variations were recorded in the protein backbone dynamics in active-site motifs of wild-type and mutants, which correlated with negligible binding energy fluctuations for the PBP3-SUL (−5.85 ± 0.04 kcal/mol) and PBP2-DUR (−5.16 ± 0.66 kcal/mol) complexes. Furthermore, higher binding affinities and low inhibition constants were noted in OXA23-DUR (−7.36 kcal/mol; 4.01 µM), OXA58-DUR (−6.44 kcal/mol; 19.07 µM), and NDM-DUR (−6.82 kcal/mol; 10.01 µM) complexes when compared with the conventional drugs avibactam and aztreonam. Stable interaction profiles of DUR with carbapenemases can possibly restore SUL activity against both PBP3_{WT} and PBP3_{MTs}. The study establishes the efficacy of the novel SUL–DUR combination as a successful treatment strategy in combating emerging XDR strains of *A. baumannii*.

Aniket Naha and Saranya Vijayakumar contributed equally to the study.

KEYWORDS

carbapenemase, molecular docking, single nucleotide polymorphisms, whole-genome sequencing, β L- β LI combination

1 | INTRODUCTION

Acinetobacter baumannii (*A. baumannii*), a nonfermenting Gram-negative coccobacillus, is responsible for several nosocomial outbreaks causing bloodstream infections, ventilator-associated pneumonia, urinary tract infections, and wound infections.^{1,2} There has been a growing concern among clinicians treating multidrug-resistant (MDR) strains of *A. baumannii*.³ Recently, the World Health Organisation (WHO) categorised carbapenem-resistant *A. baumannii* (CRAb) under priority-1: critical pathogen for research and development of new antibiotics (www.who.int). In combination with other antibiotics, colistin is currently administered for treating CRAb strains. Excessive colistin usage has resulted in increased antimicrobial resistance (AMR) and colistin-induced adverse reactions.⁴ Minocycline and tigecycline show good in-vitro activity,⁵ while cefiderocol and eravacycline are used as salvage therapy against CRAb and could potentially become first-line choices for treating *A. baumannii* infections.^{6,7} However, newer β -lactams and β -lactamase inhibitors (β L- β LI) combinations like ceftolozane-tazobactam, ceftazidime-avibactam, imipenem-relebactam, and meropenem-vaborbactam are clinically ineffective against *A. baumannii*.^{3,8}

Recently, there has been growing interest in sulbactam (SUL)-based regimens which could become a possible alternative therapeutic option. Although SUL is a β -lactamase inhibitor, it is considered as a β -lactam antibiotic due to its intrinsic activity towards both penicillin-binding protein (PBP)3 and PBP1 of *A. baumannii*.^{3,9} However, the presence of various β -lactamases could limit its use.¹⁰ Durlobactam (DUR), a novel β LI, has a diazabicyclooctane (DBO) scaffold, which inhibits a wide range of β -lactamases (primary target) and PBP2 (secondary target).⁹ Recent studies have demonstrated the potential antibacterial activity of the novel SUL-DUR (β L- β LI) inhibitor combination, which is currently under Phase-III clinical trials and may be helpful in treating infections caused by *A. baumannii*.^{9,11} Our research group has worked extensively to understand the protein-ligand interactions and complex AMR mechanisms of pathogenic bacteria through genomic analyses,^{12–15} in silico structural biology,^{16–20} and the systems biology approach.^{21–24} The present study aimed to decipher the efficacy of the SUL-DUR combination against clinical Indian isolates of *A. baumannii*. This was achieved by characterising the intermolecular binding

profiles of PBP3-SUL and β -lactamases-DUR complexes through in silico and in vitro analyses.

2 | MATERIALS AND METHODS

2.1 | Characterisation of bacterial isolates

Twenty-eight nonduplicate clinical strains of *A. baumannii* were identified from routine blood ($n = 21$) and sputum ($n = 7$) cultures from 2018 to 2019 at Christian Medical College, Vellore. All the isolates were characterised up to the species level as *A. baumannii calcoaceticus* complex (*Acb* complex) using routine biochemical and automated identification system using VITEK®-MS (BioMérieux).¹⁵

2.1.1 | Antimicrobial susceptibility testing (AST) and minimum inhibitory concentration (MIC)

Antimicrobial susceptibility testing (AST) was performed for all the isolates against 18 different antibiotics by the Kirby-Bauer disc-diffusion method and interpreted according to Clinical Laboratory Standards Institute guidelines.²⁵ The minimum inhibitory concentration (MIC) for colistin and SUL were determined for all the studied isolates using broth micro-dilution and interpreted accordingly. Two quality control (QC) strains, *Escherichia coli* ATCC 25922 and *Pseudomonas aeruginosa* ATCC 27853 were used for the colistin MIC. As an internal control, an *mcr-1* positive *E. coli* strain was also included. Additionally, two in-house QC strains, *Klebsiella pneumoniae* BA38416 (MIC 0.5 μ g/ml) and *K. pneumoniae* BA25425 (MIC 16 μ g/ml), were also included in every batch of testing. For SUL MIC, *A. baumannii* NCTC 13304 was used as a QC strain.²⁵

2.1.2 | Molecular characterisation of carbapenemase genes by multiplex-PCR

All the 28 isolates were grown on blood agar overnight, and genomic DNA was extracted using QIAamp DNA Mini Kit (QIAGEN) as per the manufacturer's

instructions. Conventional multiplex-polymerase chain reaction (PCR) was done to detect carbapenemase genes such as *bla*_{KPC}, *bla*_{IMP}, *bla*_{VIM}, *bla*_{NDM}, *bla*_{SIM}, *bla*_{OXA-48}, *bla*_{OXA-23}, *bla*_{OXA-24}, and *bla*_{OXA-58} like as described previously.¹⁴ The amplicons were visualised in 2% agarose gel stained with ethidium bromide. Known positive controls for appropriate genes were used in all of the runs (Courtesy: IHMA, Inc.).

2.1.3 | Whole-genome sequencing (WGS), assembly, and annotation

Whole-genome sequencing (WGS) was performed with the extracted DNA from all of the isolates. The quantity and quality of the DNA were determined using the Qubit 3.0 fluorometer (Thermo Fisher Scientific) and the Nanodrop spectrophotometer (Thermo Fisher Scientific). In brief, short-read sequencing was performed using IonTorrent™ Personal Genome Machine™ (Life Technologies) with 400 bp read chemistry or paired-end genomic libraries were prepared and sequenced using Illumina HiSeq V4 platform as per the manufacturer's instructions. Long-read sequencing was performed using Oxford Nanopore Technologies, using the one-dimensional sequencing method according to the manufacturer's protocol. The whole genome was obtained by performing hybrid assembly as previously described.¹³

The NCBI Prokaryotic Genome Annotation Pipeline was used to assemble and annotate all the genomes. Further downstream analysis was performed using the CGE server (<http://www.genomicepidemiology.org/>), where genomes were analysed using ResFinder v4.1 (<https://cge.cbs.dtu.dk/services/ResFinder/>) for the presence of acquired AMR genes.²⁶ The sequence type of all the isolates was assigned by the MLST v2.0 tool (<https://cge.cbs.dtu.dk/services/MLST/>) using both the Oxford and Pasteur schemes. A maximum-likelihood (ML) tree was constructed using RAxML v0.7.4²⁷ with 28 whole genomes and visualised using iTOL v6.0 (<https://itol.embl.de/>).²⁸ Each genome was annotated with the susceptibility profile of carbapenem, SUL, and colistin, the International Clone and the chromosomal mutation profile.

2.1.4 | Nucleotide accession numbers

The complete genome project of 28 studied strains were deposited at DDBJ/ENA/GenBank under accession numbers SP304 (CP040080), VB23193 (CP035672), VB473 (CP050388), VB958 (CP040040), VB1190 (CP040047), VB16141 (CP040050), VB31459 (CP035930), CIAT758

(CP038500), ACN21 (CP038644), VB35179 (CP040053), VB35435 (CP040056), VB33071 (CP040084), VB35575 (CP040087), VB2200 (CP050421), VB2486 (CP050403), P7774 (CP040259), VB82 (CP050385), VB7036 (CP050523), VB11737 (CP050400), VB723 (CP050390), PM2235 (CP050410), PM2696 (CP050412), PM3665 (CP050415), PM4122 (CP050425), PM4229 (CP050432), VB2107 (CP051474), VB2139 (CP050526), and VB2181 (CP050401).

2.1.5 | Mutation analysis of PBPs and βLs

The PBP2, PBP3, OXA23, OXA58, and NDM sequences extracted from whole genomes were scanned for nsSNPs using SeaView²⁹ and were compared with the reference *A. baumannii* ATCC17978 (CP000521). Upon nsNP detection, the stability at structural–functional aspects of the mutations was assessed with the DUET server (<http://structure.bioc.cam.ac.uk/duet>). Thermodynamic free-energy changes (ΔΔG) were computed upon site-directed mutagenesis leading to alteration in protein stability.³⁰ Further, the destabilisation effects of nsSNPs were estimated by N-H S² bond-order using the DynaMine server (<http://dynamine.ibsquares.be/>). The movement restrictions of atomic bond-vectors (S² scores) were derived directly from experimentally validated NMR chemical shifts.³¹

2.1.6 | In silico protein modelling

NCBI-BLASTp analysis against RCSB-PDB retrieved suitable templates for PBP3-3UE3, OXA23-5WI3, OXA58-4OH0, and NDM-4EY2. Global alignment (https://www.ebi.ac.uk/Tools/psa/emboss_needle/) revealed 76.1% identity of PBP3 with 3UE3. Other targets shared high identities of 98.4%, 99.6%, and 100% with OXA23, OXA58, and NDM sequences respectively. Lack of templates for PBP2, and the low (<90%) sequence identity for PBP3 resulted in the need for in silico protein modelling. Hence, extended dual-step protein modelling was performed as reported previously.¹⁸

2.1.7 | Protein structure refinement and validation

The modelled proteins (PBP2 and PBP3) and carbapenemases (OXA23, OXA58, and NDM) were scrutinised for poor-rotamers, unstable bond torsions, and outliers. These proteins were refined using the GalaxyRefine server (<http://galaxy.seoklab.org/index.html>) to enhance the structure quality by furnishing and repacking the side

chains.³² Energy minimisation was performed using Swiss-PDB Viewer v4.1.0 in vacuo with GROMOS96 43B1 force field by carrying out 2000 steps each for steepest descent and conjugate gradients.³³ The nsSNPs were introduced in wild-type (WT) protein targets to generate respective mutants (MTs) using SPDBV. The secondary protein structures were assessed using SOPMA (https://npsa-prabi.ibcp.fr/cgi-bin/npsa_automat.pl?page=npsa_sopma.html), and tertiary structures were validated with ProSA-web,³⁴ HARMONY,³⁵ and ProTSAV³⁶ servers. UCSF CHIMERA was used for protein structure visualisation.³⁷

2.1.8 | Active site prediction

Several reports highlighted three conserved motifs (S-X-X-K, S-X-N/D, and K-T/S-G-T) within PBPs playing pivotal roles in β L binding.^{38,39} The S-X-X-K motif in β -lactamases was conserved with PBP2⁴⁰ while active-sites of NDM were retrieved from literary evidence⁴¹ and validated through CASTp-3.0 server (<http://sts.bioe.uic.edu/castp/calculation.html>).⁴²

2.1.9 | Ligand selection

The ligands for the present study were procured from the PubChem Database (<https://pubchem.ncbi.nlm.nih.gov/>). The three-dimensional coordinates of SUL (130313), DUR (89851852), avibactam (AVI) (9835049), and aztreonam (AZT) (5742832) were generated using the CORINA server (https://www.mn-am.com/online_demos/corina_demo). The polar H-atoms were added, and spatial arrangements of ligands were visualised in Avogadro software v1.2.0.⁴³

2.1.10 | Molecular docking analysis

Refined proteins were subjected to molecular docking analysis with AutoDock 4.2. Both the WT and MT protein targets were optimised before docking analysis by adding polar H-atoms, Kollman charges, and merging non-polar H-atoms thus stabilising the free end residues of the protein moiety. The ligand torsions were fixed upon the addition of Gasteiger charges. The affinity grid box was constructed centring the active-site residue coordinates of dimensions $60\text{\AA}^3$ with equal spacing of $0.375\text{\AA}$. Docked poses were generated based on the Lamarckian and genetic algorithms, while the protein-ligand complex with least binding energy (BE) was selected for the present study.⁴⁴ The docked complexes depicting several intermolecular interactions were

visualised in Discovery Studio 2020 v20.1.0.19295.⁴⁵ Molecular docking of SUL was performed with PBP3 while DUR was docked with β -lactamases and PBP2. The classical drugs AVI and AZT were positive controls against DUR- β -lactamase and DUR-NDM complexes respectively.

3 | RESULTS AND DISCUSSION

3.1 | Bacterial isolates and antimicrobial susceptibility

All 28 *Acb* complexes were confirmed as *A. baumannii* using VITEK®-MS and re-confirmed by the presence of an intrinsic *bla*_{OXA-51} like gene by uniplex-PCR. AST revealed six isolates as pan-drug resistant (PDR) being resistant to all the tested antibiotics while 21 showed XDR profiles being resistant to all the tested antibiotics except for minocycline and tigecycline (blood, $n = 20$ and sputum, $n = 06$). Of the remaining two isolates, one from Blood was MDR (PM4229) while another isolate (SP304) from sputum was pan-drug susceptible (PDS) (Supplementary File S1]. All the isolates ($n = 28$) were SUL resistant. Five isolates (blood, $n = 04$; sputum, $n = 01$) were colistin intermediate, while the remaining 23 isolates were colistin susceptible (blood, $n = 17$; sputum, $n = 06$). The SUL and colistin MIC range, MIC₅₀, and MIC₉₀ of the blood and sputum isolates were tabulated (Table 1).

3.2 | Carbapenemase PCR and WGS

Multiplex-PCR for carbapenemase genes showed the predominance of class-B and D carbapenemase, *bla*_{OXA-23}, and *bla*_{NDM} like genes (Figure 1). Only two isolates were positive for *bla*_{OXA-58} like. WGS revealed eight variants of intrinsic *bla*_{OXA-51} gene (*bla*_{OXA-66}, *bla*_{OXA-64}, *bla*_{OXA-68}, *bla*_{OXA-94}, *bla*_{OXA-144}, *bla*_{OXA-203}, *bla*_{OXA-337}, and *bla*_{OXA-371}), two variants of *bla*_{OXA-23} (*bla*_{OXA-23} and *bla*_{OXA-169}), two variants of *bla*_{OXA-58} (*bla*_{OXA-58} and *bla*_{OXA-420}), and a single variant of *bla*_{NDM} (*bla*_{NDM-1}).

TABLE 1 MIC range, MIC₅₀, and MIC₉₀ of clinical strains of *Acinetobacter baumannii* tested against sulbactam and colistin ($n = 28$)

	Sulbactam ($\mu\text{g/ml}$)	Colistin ($\mu\text{g/ml}$)
MIC range	8–256	0.03 to >16
MIC ₅₀	256	1
MIC ₉₀	256	>16

Abbreviation: MIC, minimum inhibitory concentration.

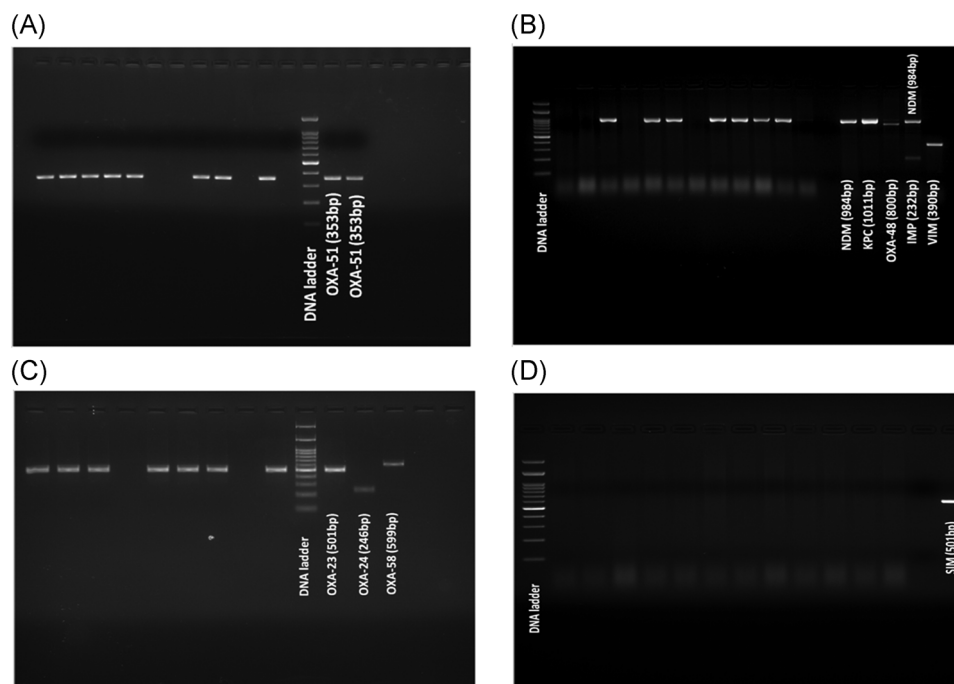


FIGURE 1 Gel electrophoresis of carbapenem-resistant genes in *Acinetobacter baumannii*: (A) Intrinsic *bla*_{OXA-51} gene. (B) Different class of carbapenem-resistant genes. (C) Class D *bla*_{OXA-23}, *bla*_{OXA-24}, and *bla*_{OXA-58} genes. (D) Class B *bla*_{SIM} gene

Fourteen isolates carried *bla*_{OXA-23} alone, 10 isolates co-harboured *bla*_{OXA-23} and *bla*_{NDM-1}, one was a coproducer of *bla*_{OXA-58} and *bla*_{OXA-23} while another cocarried *bla*_{OXA-420} and *bla*_{NDM-1}. The remaining two isolates were identified as carbapenemase negative. Based on Oxford and Pasteur's MLST schemes, 28 isolates were categorised into four International clonal lineages (ICs) and one clonal complex (CC). Four novel STs were identified, while IC2^{Oxf}/CC2^{Pas} was the predominant clone in 16 isolates, followed by IC8^{Oxf}/CC10^{Pas} in four, IC7^{Oxf}/CC25^{Pas} in two, IC1^{Oxf}/CC1^{Pas} and CC85^{Oxf} in one isolate each. The ML phylogenetic tree from 28 whole genomes was mapped to the reference genome *A. baumannii* ATCC 17978 (CP012004) (Figure 2).

3.3 | Screening of nsSNPs from target proteins

PBP2_{I108V} (17.85%) was the most prevalent mutant allele, while PBP2_{G264S} and PBP2_{S259T} were found in one isolate each. Amongst PBP3_{MTs}, the predominant mutant allele was PBP3_{A515V} (35.71%) followed by PBP3_{T526S} (10.71%). The in vitro assay revealed an increment in SUL MIC (16–256 µg/ml) with either of these PBP3_{MTs}. Three isolates (VB473, VB1190, and VB7036) with dual carbapenemase (*bla*_{OXA-23}, *bla*_{NDM-1}) had a MIC of 256 µg/ml. All PBP3_{A515V} isolates belonged to the IC2^{Oxf}/CC2^{Pas} clone, while 80% of PBP2_{I108V} isolates belonged to the IC8^{Oxf}/

CC10^{Pas} lineage. No mutations were detected in OXA23 and NDM, whereas the CIAT758 strain possessed A256D mutation in OXA58. Both OXA23 and NDM were flanked with insertion sequence (IS) elements IS_{Aba1} (Tn2006) and IS_{Aba125} (Tn125), respectively. These composite transposons facilitated easy spread and efficient transmission resulting in a high prevalence of OXA23 and NDM in the isolates. Though IS_{Aba3} flanks OXA58, the presence of a transposon has not been reported so far, thereby lowering its prevalence and spread.⁴⁶ The backbone dynamics of the neighbouring residues (amino-acid: 239–272) of OXA58_{A256D} revealed a 0.83% reduction in backbone rigidity compared with the OXA58_{WT}. Due to the flexible backbone dynamics, the enzymatic degradation capability of OXA58_{MT} was enhanced, which may be a significant reason for gaining A256D mutation in OXA58.

The negative $\Delta\Delta G$ values from the DUET server signified the destabilising effects of nsSNPs at structural–functional levels. The extent of destabilisation was further validated with N–H S² bond-order restraints (Supplementary File S2A). Residue-level analysis of PBP2_{I108V} (S²-score = 0.88) and PBP2_{S259T} (S²-score = 0.9) revealed rigid backbone conformers while PBP2_{G264S} had a relatively flexible backbone (S²-score = 0.78) when compared with PBP2_{WT}. The PBP2_{G264S} mutation ($\Delta\Delta G = -1.2$ kcal/mol) had the lowest $\Delta\Delta G$ scores amongst PBP2_{MTs} and therefore, showed a destabilisation profile leading to protein unfolding at low energy

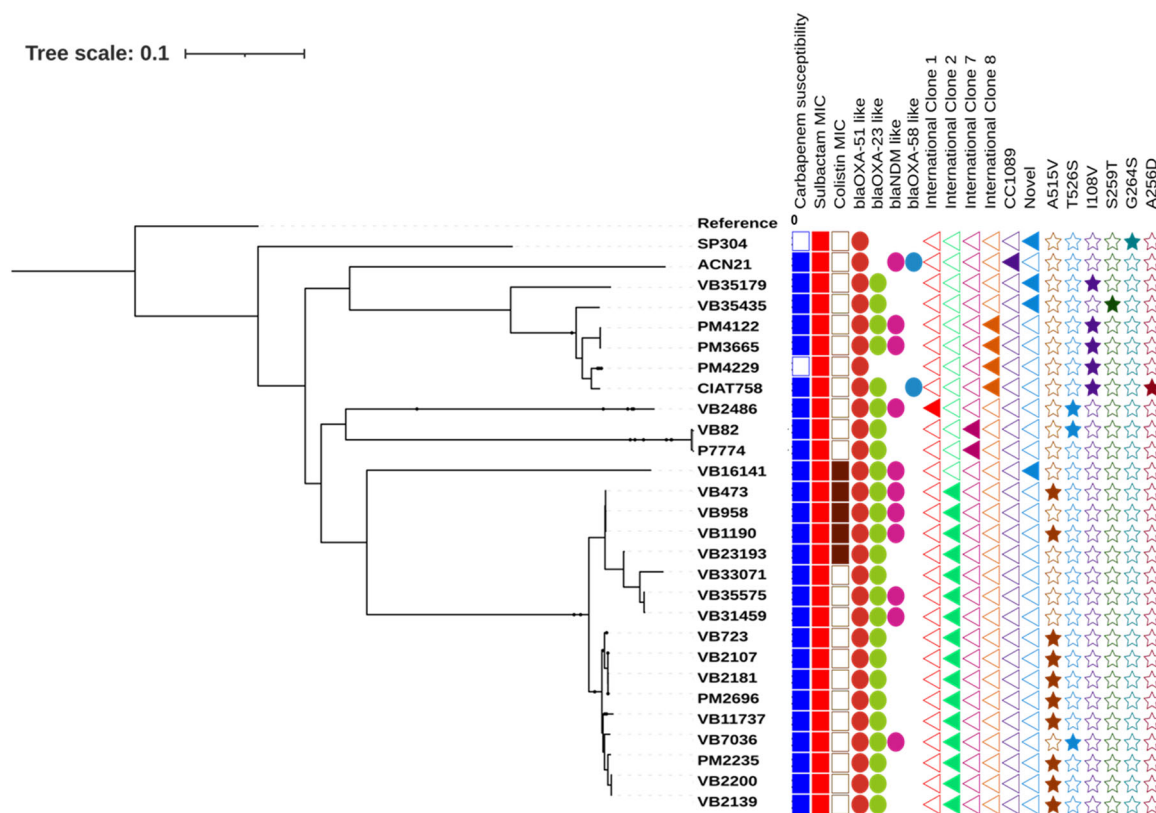


FIGURE 2 A maximum-likelihood phylogenetic tree of 28 whole genomes sequenced in this study. The colour-filled boxes under carbapenem and sulbactam minimum inhibitory concentration (MIC) indicated resistant, while for colistin MIC they indicated intermediate, and all the empty boxes indicated susceptible. The colour-filled shape denoted presence while the empty shape denoted absence of the respective traits

enforcements. Similarly, PBP2_{I108V} (−1.123 kcal/mol) and PBP2_{S259T} (−0.476 kcal/mol) correlated with the most stable backbone dynamics. The PBP3_{T526S} mutation resulted in protein flexibility (S^2 -score = 0.63), while reduced flexibility (S^2 -score = 0.8) was observed in the PBP3_{A515V} mutation. The backbone stability profiles established the lower energy requirement for protein unfolding upon PBP3_{T526S} (−1.45 kcal/mol) compared to the PBP3_{A515V} (0.34 kcal/mol) mutation. Thus, the protein backbone dynamics validated the free-energy changes upon mutations and their effects on protein stability.

3.4 | In silico structure prediction, refinement, and validation

The five protein targets were refined and subjected to multiple validations. The validated modelled structure of PBP2 and PBP3 were submitted to the Protein Model Database (PMDb) (<http://srv00.recas.ba.infn.it/PMDB/>), having PMDB-IDs PM0084111 and PM0084112, respectively. Lower MolProbity scores, minimum percentages of poor rotamers, and Ramachandran favoured zones

(>95%) defined stable conformations for all the protein targets with reduced steric clashes and outliers. The protein structures and the active sites predominantly constituted of random coils, and had a solvent accessibility area of $76.86 \pm 1.4\%$, while $23.94 \pm 6.8\%$ was buried (Supplementary File S3). Global Z-scores from the ProSA-web signified that the target proteins were energetically stable as they fell within the defined zones of experimentally characterised proteins (Figure 3A). HARMONY propensity calibration plots revealed that all the targets lay extremely close to the straight fit line, revealing negligible gross errors in folding patterns (Figure 3B). The ProTSAV heat maps revealed a unified Z-score, confirming protein structure quality within 2–5 Å RMSD, making them ideal for protein-ligand interaction studies³⁶ (Figure 3C).

3.5 | Active-site validations in target proteins

The three conserved motifs S-X-X-K, S-X-N/D, and K-T/S-G-T, were present in both PBP2 and PBP3 as [S326-T327-I328-

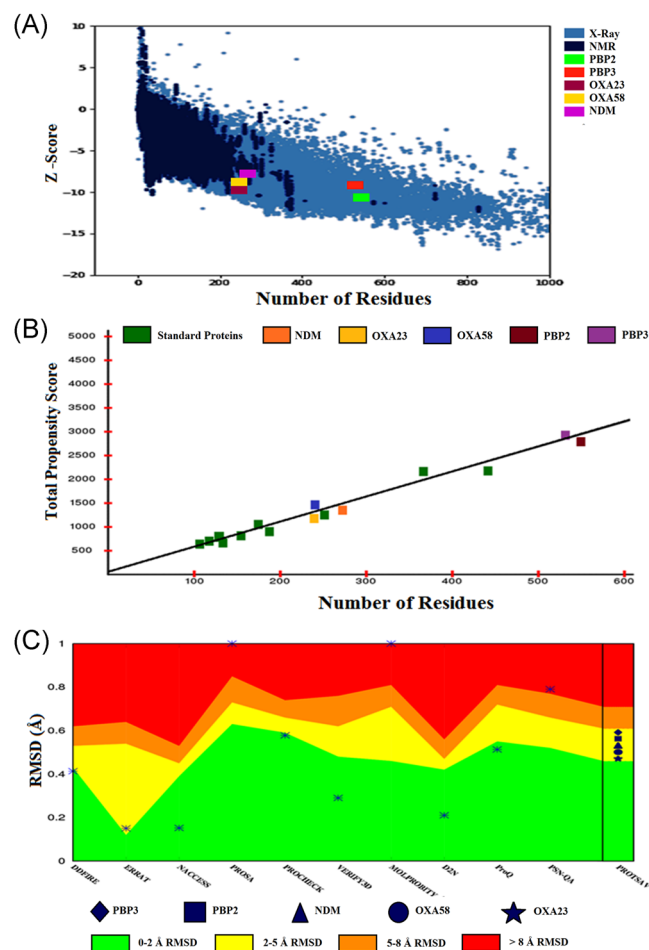


FIGURE 3 Structure evaluations of modelled proteins: (A) Global model quality of protein targets. (B) Propensity plots of protein targets. (C) ProTSAV heat-maps of protein targets

K329] and [S336-T337-M338-K339]; [S383-C384-D385] and [S390-S391-N392]; and [K537-T538-G539-T540] and [K525-T526-G527-T528], respectively. The S-X-X-K motifs for OXA23 [S79-T80-F81-K82] and OXA58 [S83-T84-F85-K86] were identified along with other key residues of OXA23 (F110 and M221) and OXA58 (F114 and M225). In NDM, the residues involved in the substrate binding (N220 and D124) were considered as active sites as previously reported.¹⁹ Active-site motifs/residues of all the proteins targets were validated with the CASTp server before performing molecular docking analysis.

3.6 | Molecular docking of PBPs and β -lactamases with β L- β LI combination

The variation in BEs of PBP3-SUL and PBP2-DUR complexes was analysed and correlated with the N-H S^2 bond-order restraints to decipher the effect of nsSNPs on the binding profiles of the complexes. The BEs of protein-ligand complexes are represented in Figure 4. There were no variations among the backbone dynamics of STMK (average $S^2 = 0.71$) and SSN (average $S^2 = 0.77$) motifs of PBP3_{WT} and PBP3_{MTs} (Supplementary File S2B). PBP3_{T256S} (average $S^2 = 0.65$) and PBP3_{A515V} (average $S^2 = 0.7$) lay within and in close vicinity of the KTGT motif respectively, resulting in flexible protein backbone dynamics. However, there were no significant differences in BEs (-5.85 ± 0.04 kcal/mol) and inhibition constants (51.36 ± 3.7 μ M) of PBP3-SUL complexes. The flexible S336 residue ($S^2 = 0.71$) facilitates motif deprotonation followed by a nucleophilic

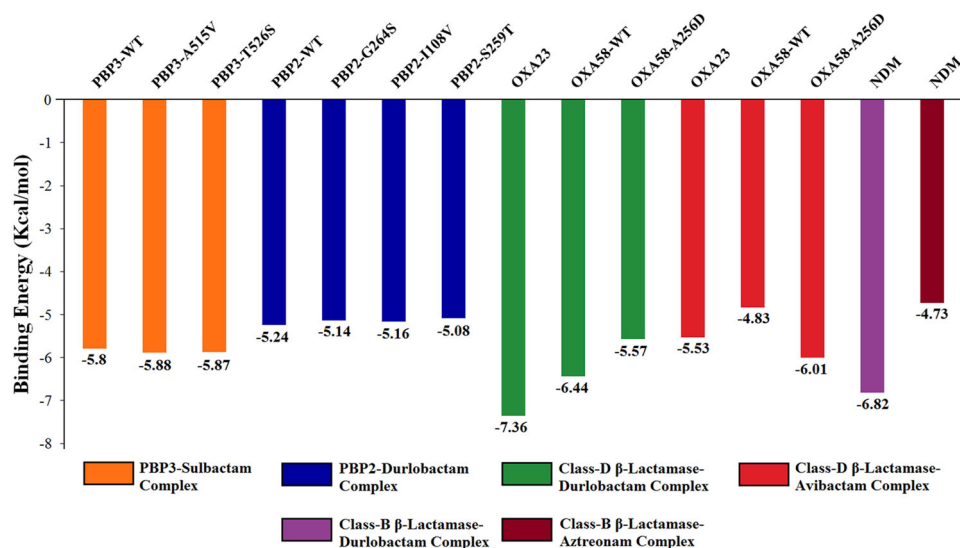


FIGURE 4 Binding energies of docked complexes

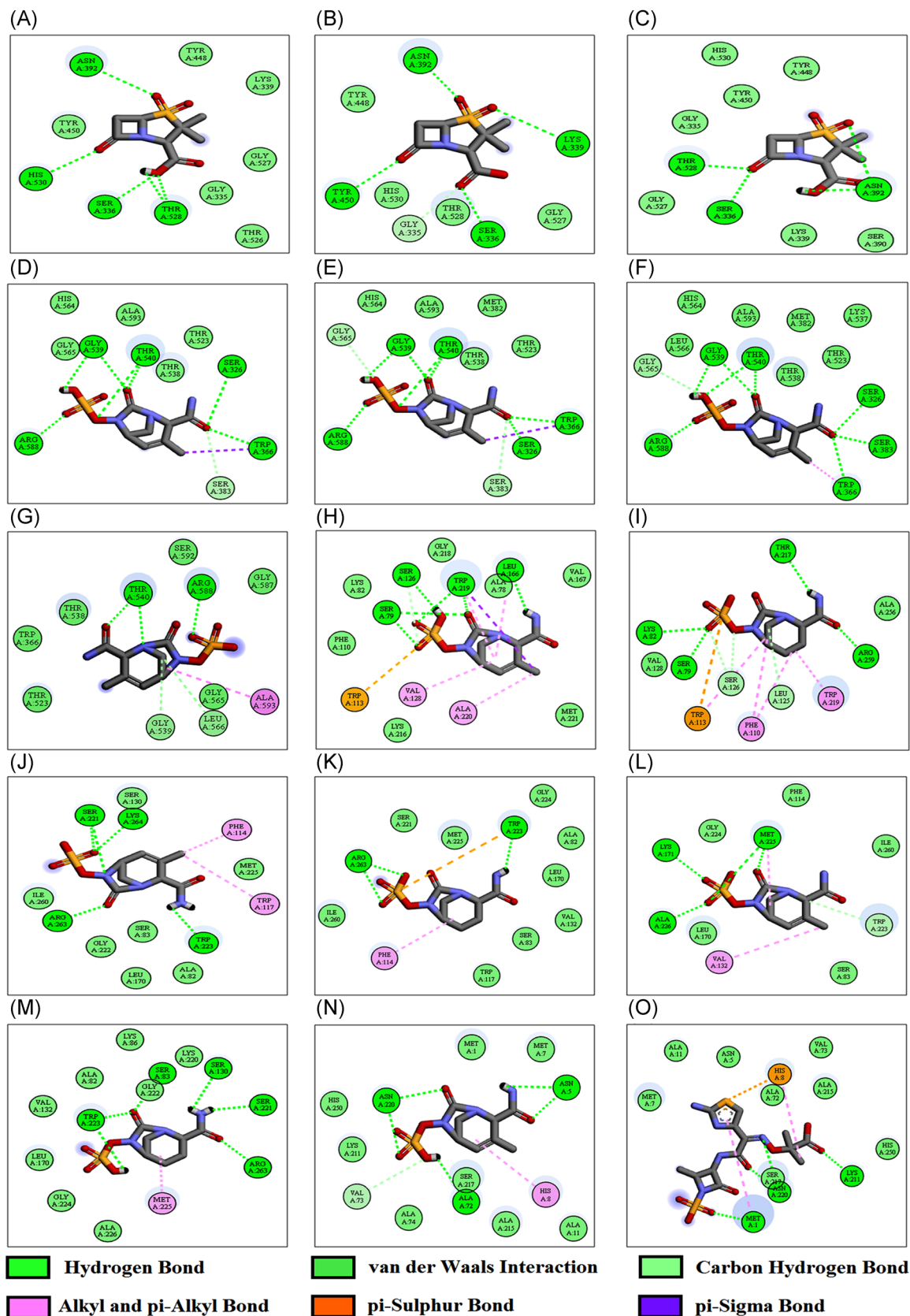


FIGURE 5 Docked poses of different protein-ligand complexes: (A) PBP3_{WT}-SUL; (B) PBP3_{T526S}-SUL; (C) PBP3_{A515V}-SUL; (D) PBP2_{WT}-DUR; (E) PBP2_{G264S}-DUR; (F) PBP2_{I108V}-DUR; (G) PBP2_{S259T}-DUR; (H) OXA23-DUR; (I) OXA23-AVI; (J) OXA58_{WT}-DUR; (K) OXA58_{WT}-AVI; (L) OXA58_{A256D}-DUR; (M) OXA58_{A256D}-AVI; (N) NDM-DUR; (O) NDM-AZT. AVI, avibactam; AZT, aztreonam; DUR, durlobactam; PBP, penicillin-binding protein; SUL, sulbactam

attack through the flexible K339 residue ($S^2 = 0.75$) on the amide carbonyl-carbon of the β L-ring forming a tetrahedral intermediate. Flexible residues of STMK and KTGT motifs stabilised the tetrahedral intermediate by stable H-bonds with the carbonyl O-atom of β L-ring inside the oxyanion hole (Figure 5A–C). Finally, the tetrahedral intermediate collapses, expelling the N-atom of β L-ring. A stable acyl-enzyme intermediate was thereafter formed, assuring proper inhibition of the target protein upon protonation by the S390 residue (SSN motif).³⁸

In PBP2, though individual residue-level analysis of the nsSNPs revealed minimal fluctuations altering backbone flexibilities, there was no change in S^2 values in any of the PBP2 active-site motifs (Supplementary File S2B). The observation may be a crucial factor determining similar BEs (-5.16 ± 0.66 kcal/mol) of PBP2–DUR complexes. An approximate 20% increase in the inhibition constant was recorded for PBP2_{MTs} when compared with PBP2_{WT}. The PBP2–DUR interaction incorporated the S326 and K329 residues of STIK (average $S^2 = 0.75$), and the G539 and T540 residues of KTGT (average $S^2 = 0.72$) motifs forming stable H-bond interactions with the carbonyl O-atom of the DBO ring inside the oxyanion hole (Figure 5D–G).

OXA23–DUR (-7.36 kcal/mol) displayed a better interaction profile than OXA23–AVI (-5.53 kcal/mol). The carbonyl O-atom of the DBO ring in DUR was positioned inside the oxyanion hole, forming an H-bond with S79 of STFK motif, which was not observed with the carbonyl O-atom of AVI. The OXA23–DUR complex was further stabilised with F110 and M221, forming vdW interactions from either side of the active-site groove. The aromatic π -ring of W113 formed a π -S interaction with S-atom of DUR while W219 formed a π - σ interaction providing additional stability to the OXA23–DUR complex (Figure 5H,I). Similarly, the binding affinity of OXA58_{WT}–DUR (-6.44 kcal/mol) was higher than that of OXA58_{WT}–AVI (-4.83 kcal/mol). The OXA58_{WT}–DUR complex was further stabilised by vdW interactions with S83 and M225, and a π -alkyl interaction with an aromatic F114 residue. On the contrary, OXA58_{A256D}–AVI possessed a slightly better-docked profile (-6.01 kcal/mol) than OXA58_{A256D}–DUR (-5.57 kcal/mol). Analysing the docked poses of OXA58_{A256D}–AVI (6 H-bonds) revealed H-bond interaction inside the oxyanion hole with a carbonyl O-atom of the DBO ring in AVI with S83. The MT-complex was further stabilised with a π -alkyl interaction of the DBO ring with the M225 residue. The OXA58_{A256D}–DUR complex was stabilised by four H-bonds and vdW interactions with the active-site residues (S83 and F114). Mutation in OXA58 was rarely encountered⁴⁶; therefore though AVI has shown better profiles than DUR, the efficacy of DUR cannot be

ruled out as the BEs and interactions of the OXA58_{A256D}–DUR complex was relatively stable (Figure 5J–M). Molecular docking analysis revealed better-docked profiles for NDM–DUR (-6.82 kcal/mol) than NDM–AZT (-4.73 kcal/mol). Though both the ligands bound to N220, the carbonyl O-atom of the DBO ring in DUR was positioned in the oxyanion hole, thereby ensuring better stability of the NDM–DUR complex (Figure 5N,O). The intermolecular interactions of SUL and DUR with their respective targets displayed strong binding affinities as compared to the classical drugs AVI and AZT. The present study thus established the potency of the SUL–DUR combination against both the MT and WT strains of MDR *A. baumannii*.

4 | CONCLUSION

The recent emergence of XDR nosocomial strains of *A. baumannii* has created significant concern due to the extreme difficulty in their effective treatment. Their AST has demonstrated alarming XDR profiles. The CRAB strains have prompted the shift to newer β L– β LI inhibitor combinations to combat the resultant AMR. WGS and PCR analysis in the current study has revealed the presence of several Class-B and D β -lactamases resulting in 100% SUL resistance in all of the studied Indian isolates. The binding affinities and inhibition constants of DUR against different β -lactamases predict its multi-target specificity. The current study also hypothesises that DUR may be highly effective in treating strains harbouring NDM in combination with other carbapenemases. For the treatment of OXA58_{A256D} containing strains, a higher dosage of DUR may be required to achieve effective inhibition, although encouraging further in vivo validations. The combination therapy of SUL–DUR could restore SUL activity, as evident from its significant binding affinities towards both PBP3_{WT} and PBP3_{MTs}. Overall the study has displayed substantial evidence for establishing the efficacy of SUL–DUR, a novel β L– β LI combination to combat the pathogenicity of CRAB strains.

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CONFLICT OF INTERESTS

The authors declare that there are no conflict of interests.

AUTHOR CONTRIBUTIONS

Aniket Naha: Data curation, in silico methods and analysis, writing-original draft. **Saranya Vijayakumar:** In vitro methods-WGS analysis and writing-original draft. **Binesh Lal:** Manuscript correction and supervision. **Baby Abirami Shankar** and **Suriya Chandran:** Isolate revival, PCR, in vitro MIC, and AST. **Sudha Ramaiah:** Validation of methodology, software, and manuscript review. **Anand Anbarasu** and **Balaji Veeraraghavan:** Conceptualisation, project administration, funding acquisition, and manuscript review.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

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SUPPORTING INFORMATION

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